Document Control

Title

Continuous Positive Airway Pressure (CPAP) and Bi-level Continuous Positive Airway Pressure (BiPAP) Neonatal Nursing Care Guideline

Author	**Author's job title** Senior Neonatal Nurse
Directorate **Clinical Support & Specialist Services**	**Department** Special Care Baby Unit

Version	Date Issued	Status	Comment / Changes / Approval
1.0	2009	Final	Approved for use at Northern Devon Healthcare NHS Trust.
1.1	2013	Revision	Revised
1.2	2014	Revised	Revised and guidelines for BiPAP added.
2.0	2015	Final	Approved by Paediatric Specialty Team 27/02/15
3.0	Feb 2019	Final	Approved by Paediatric Speciality Team 01/03/19
4.0	Feb 2022	Final	Approved by Paediatric Speciality Governance 18th Feb 2022

Main Contact Special Care Baby Unit North Devon District Hospital Raleigh Park Barnstaple, EX31 4JB	**Tel: Direct Dial** – **Tel: Internal** – **Email**:

Lead Director
Director of Nursing

Superseded Documents
Continuous Positive Airway Pressure (CPAP) and Bi-level Continuous Positive Airway Pressure (BiPAP) Neonatal Nursing Care Guideline v3.0

Issue Date	**Review Date**	**Review Cycle**
Feb 2022	Feb 2025	Three years

Consulted with the following stakeholders: (list all)

- Previously consulted and approved by South West Neonatal Nurses Benchmarking Group
- Neonatal Nurses
- Paediatricians

Approval and Review Process

- Peadiatric Speciality Team

Local Archive Reference
G:\paediatric resources\neonates\neonatal guidelines\previous versions of guidelines
Local Path
G:/Paediatric Resources\Neonates\Neonatal guidelines
Filename

CPAP and BIPAP Neonatal Nursing Care Guidelines v4.0	
Policy categories for Trust's internal website (Bob) Neonatal	**Tags for Trust's internal website (Bob)** Neonate, cpap, bipap, respiratory, grunting, respiratory distress syndrome, CO2

CONTENTS

Document Control 1
1. Purpose 3
2. Definitions 3
3. Responsibilities 4
Role of the neonatal nurse designation 4
Role of the medical team 4
4. Background 4
5. Contraindications 5
6. Equipment 5
7. Nursing care 5
8. Nursing observation/ documentation 8
9. Discontinuing CPAP 8
10. Biphasic level positive airway pressure 9
11. Monitoring Compliance with and the Effectiveness of the Guideline 10
Standards/ Key Performance Indicators 10
Process for Implementation and Monitoring Compliance and Effectiveness 10
12. References 10
13. Associated Documentation 13
14. Appendices 14
15. Skin assessment score for pressure areas specific to SiPAP and Ventilated infants 14
Associated Documentation (Optional) 14

1. Purpose

1.1. The purpose of this document is to detail the process for providing respiratory support by use of continuous positive airway pressure (CPAP) and Bi-level continuous positive airway pressure (BiPAP).

1.2. The policy applies to all staff involved in the care of neonates.

1.3. Implementation of this policy will ensure that:

- We can support the safe care of a neonate requiring respiratory support in the form of CPAP and BiPAP
- The risks associated with the management of a neonate requiring respiratory support are reduced

2. Definitions

2.1. BiPAP - Bi-level continuous positive airway pressure

2.2. CPAP - Continuous Positive Airway Pressure

2.3. FRC - Functional Residual Capacity

2.4. NCPAP - Nasal Continuous Positive Airways Pressure

2.5. NIPPV - Nasal intermittent Positive Pressure Ventilation

2.6. MDT - Multidisciplinary Team

2.7. PEEP - Positive End Expiratory Pressure

2.8. PIP - Peak Inspiratory Pressure

3. Responsibilities

Role of the neonatal nurse

3.1. The neonatal nurse is responsible for:

- Keeping the Infant Flow SiPAP machines in readiness for use
- Ensuring that settings and the maintenance of the Infant Flow SiPAP machine are in accordance with medical team instructions
- Keeping medical team aware of infants condition
- Observing the baby for associated risks and taking appropriate action
- On-going monitoring and care of the baby supported by CPAP/ BiPAP

Role of the medical team

3.2. The medical team is responsible for:

- Acting in accordance with this guideline
- Ensuring that risks associated with CPAP/BiPAP use are minimised
- Keeping the Tertiary unit updated on condition of infant and delivering collaborative plan of care.

4. Background

4.1. Continuous Positive Airway Pressure (CPAP) is the application of positive pressure to the airway of a spontaneously breathing infant throughout the respiratory cycle (Lavelle and McCoy, 2018).

4.2. CPAP works by splinting the upper airway & reducing obstructive apnoea, preventing alveolar collapse, conserving endogenous surfactant and reducing the respiratory rate (Gupta and Donn, 2016).

4.3. CPAP can be used to help treat:

- Respiratory Distress Syndrome (RDS)
- Apnoea of Prematurity
- Transient Tachypnoea of the Newborn (TTN)
- Infants with Chronic Lung Disease

- Prevention of extubation failure

4.4. Babies requiring CPAP, BIPAP but not fulfilling any other criteria for intensive care are classed as High Dependency, (BAPM, 2015). All babies receiving CPAP/ BiPAP should be discussed with the tertiary unit at the earliest opportunity and a collaborative care plan made.

5. Contraindications

- Known pneumothorax
- Facial and nasal abnormalities e.g. choanal atresia, cleft palate, tracheoesophageal fistula (TOF)
- Diaphragmatic hernia
- Larger babies often do not tolerate application of CPAP

6. Equipment

- SIPAP Flow driver, circuit and generator
- Humidity System
- Bag of sterile water
- Appropriate size headgear using manufacturers sizing tool
- Appropriate size prongs/ mask using manufacturers sizing tool
- Relevant monitoring equipment – will need minimum of O2 saturation monitor. Full cardiac monitoring may be appropriate dependent on infant's condition.
- Observation chart
  Gastric tube (and drainage bag/ syringe with plunger removed to vent air from stomach)

7. Nursing care

7.1. The circuit should be set up and attached to the infant following the step by step guide given in the SIPAP Infant Flow Driver Manual.

7.2. Humidification

Humidification and warming of the inspiratory gases is very important. High flow gases dry the mucosa, decrease the muco-ciliary function and increase the airway resistance (Lavelle and McCoy, 2018).

- CPAP circuits should be humidified at all times.
- Ensure that the humidifier is on and warmed prior to connecting to an infant. Failure to do this may lead to a cold infant.
- Sterile water is used to fill humidification chambers.
- Don't overfill the humidification chamber, don't let it run dry.
- Humidification is delivered to the infant at 37°c, set to invasive mode.

7.3. Headgear

- Ensure the correct size hat is used. Measure the infants head circumference from crown at back of the head to the brow line and select the corresponding size hat.

- The headgear size should be reviewed regularly, and changed when dirty or ill fitting.

7.4. Nasal Prongs

- Measure the nasal prongs/ mask size using the diagram in each CPAP generator pack. They are colour co-ordinated for small/ medium/ large.
- Attach the prongs/mask to the generator prior to attaching to the infant.
- Prongs should be positioned correctly so that they are square onto the nose and not tipped at an angle. They should NOT be pressed hard against the nasal septum (this is to prevent skin damage)
- Masks should be place evenly on the nose to create a seal and minimise gas leak especially to the eyes.
- Prongs/mask should be kept clean, patent and free from obstruction.
- Alternate between prong and mask when the infant tolerates this.
- Correct fitting of the prongs/mask, observation and treatment should be documented in nursing records. See Skin Integrity Care Plan, skin care guideline, and appendix one for assessment tool.

7.5. Respiratory

The required positive pressure varies greatly depending on the infant's condition. Initial PEEP is usually 6-8cmH2O (Jollye and Summers, 2010), however, in certain instances a consultant may ask for PEEP of up to 10cmH2O. High pressure over 10cmH2O may restrict pulmonary blood flow, increase air leak risk, and cause over-distension and later hypercapnia.

Pressure is achieved by altering the flow. The usual starting rate is 8L/min. Aim to keep flow to a minimum whilst ensuring adequate respiratory support is being delivered.

Decision on pressures and flow are to be directed by the medical team.

Oxygen requirements should initially be set to the same level the infant is requiring via alternative methods and then adjusted accordingly.

Blood gases should be checked one hour post commencement of CPAP then under medical guidance and dependent on infant's condition. Generally this can be 4-6 hrly initially decreasing as condition stabilises.

7.6. Nutrition

- Initially babies will tend to be nil by mouth and will require intravenous fluids. Enteral feeding is not contra-indicated for babies receiving CPAP. Introduction of milk feeds by any method should always be carried out following liaison with medical staff and a plan should be devised on an individual basis depending upon the infant's ability to tolerate enteral feeds and overall condition. Please refer to Enteral Nutrition in Neonates Guideline, Newborn Infant Feeding Policy and Neonatal Care Planning Standard Operating Procedure.

- An oral gastric tube should be used whenever possible and aspirated prior to every feed to avoid 'CPAP belly'. Use of a nasal gastric tube should be avoided for 3 main reasons: It increases the work of breathing, can interfere with obtaining a good seal and may increase nasal irritation.

7.7. Comfort

- Minimal handling is essential for the sick neonate (See Developmental Care guideline). 'Hands on' intervention should be limited and cares clustered.
- Good fixation and positioning will enhance infant's comfort. Babies can be nursed prone or supine when receiving NCPAP and may also have kangaroo care (see developmental care guidelines).
- Elevate cot to 30° - Elevated cot will reduce pressure on the neonates diaphragm

If the infant does not settle try the following methods:-

- Check the prongs/mask are positioned correctly and are comfortable
- Check the airway is clear, if there a buildup of secretions, the infant may need suctioning
- Try usual calming methods e.g. containment, nesting, swaddling, pacifier etc (see guidelines for developmental care, and non-nutritive sucking).
- Aspirate any excess gastric air via orogastric tube
- Try nursing the infant prone to relieve abdominal distension, diaphragmatic pressure, and aid air entry
- Consider pain relief, (see pain management guidelines).
- Kangaroo care or skin-to-skin with parents/ carers can be encouraged to aid regulation of respiration – this can be assessed on an individual basis.
- Water vapour condenses and gathers in the CPAP tubing, this needs to be removed as soon as it is noticed, because it is very noisy for the infant when it swings in the tubing and can also move and blow into the infant's nose causing distress and affecting breathing.

7.8. Parental involvement

- Give parents information about their infant's condition

Encourage involvement in their infant's care. Nurses should facilitate family integrated care and parents should be encouraged to participate in their infants care and encouraged to have skin to skin contact when the infants condition allows.

8. Nursing observation/ documentation

- Infant's condition, care given and evaluation should be documented in the nursing records.
- Observe and document a baseline assessment of the infant prior to the commencement of NCPAP and hourly thereafter. Including;-
  - Respiration – rate, colour, effort, breath sounds, signs of distress (tachypnoea, nasal flaring, recessions, grunting). Oxygen requirements and saturation levels
  - Temperature
  - Cardiovascular – heart rate, central and peripheral circulation, blood pressure
  - Neurological – tone, irritability, signs of pain/distress.
  - The time NCPAP is commenced needs to be documented in the nursing record along with the prong and hat size and the prescribed pressure.
  - Skin integrity (see appendix one)
- Hourly recordings of the measured level of PEEP should be made, (this should equal prescribed level). NB if the measured and prescribed level of pressure is not equal DO NOT automatically tighten fixation straps but check the system to ensure there are no leaks. Occasionally pressure will not match due to the infant keeping its mouth open and air escaping. Dummies or pacifiers may be utilised" (see non-nutritive sucking guideline) however informed consent should be sought from parents/ carers before their use.

9. Discontinuing CPAP

Signs baby may be ready to try time off CPAP, or have CPAP pressure weaned	Signs baby is not ready to try time off CPAP or have pressure weaned
Having infrequent or no desaturations	Having frequent or profound desats
Having infrequent or no bradycardias	Having frequent or profound bradycardias
Effort of breathing minimal	Baby persistently tachyapnoeic
Blood gas stable	Recession apparent when breathing observed
Oxygen requirement stable and less than 30%.	Oxygen requirement greater than 30% and/or rising.
Baby has tolerated brief times off CPAP, ie when weighed or during cares.	Baby does not tolerate brief time off CPAP when weighed or having cares done.
Baby copes well with handling	Baby does not cope with handling well.

10. Biphasic level positive airway pressure

Bi-level pressure respiratory support. Also known as Biphasic CPAP

- PIP is delivered at predetermined intervals/ synchronised intervals (depending on mode) above CPAP/ PEEP at a pressure and rate set by the medical team. Small incremental pressure increases of 2 to 3cmH2O augment functional residual capacity and can offload respiratory work of breathing. BiPhasic mode has been shown to improve oxygenation and ventilation when compared to CPAP (Carefusion, 2018)

10.1. BiPhasic mode

- BIPAP is time triggered as opposed to patient triggered. The SIPAP machine will ensure a set amount of pressure supported breathes are delivered within a minute.
- PIP is delivered intermittently at predetermined time intervals above the baseline CPAP/ PEEP. Target pressure and rate are set by the medical team.
- Not synchronized with respiratory effort. This mode is better suited to patients with a poor respiratory drive as can provide adequate support. Rate needs to commence at same level as babies own respiratory drive, and be altered accordingly.
- The SIPAP machine is able to provide pressure supported breaths in the event of an apnoeic episode as long as all set up settings have been confirmed.

10.2. BiPhasic trigger mode

- A non-invasive synchronized nasal intermittent positive pressure ventilatory (SNIPPV) support modality that allows clinicians to provide respiratory support without an endotracheal tube (Carefusion, 2018).
- PIP is delivered above CPAP/ PEEP at a pressure set by the medical team in synchrony with the neonate's respiratory effort (sensed through an abdominal sensor). Each time the patient initiates a breath the SIPAP machine will deliver a fully supported breath.
- Inspiratory time (Ti) and back up rate can be varied. Back up rate function is only triggered if the neonate becomes apnoeic or spontaneous breaths are less than back up rate. As there is a need for the patient to actively trigger a breath this mode is better suited to patients who have an adequate respiratory drive.

10.3. For all Biphasic modes

8 litres should deliver approximately 5cm of pressure, this will depend on a good seal. Clinician set to deliver a flow on top of the baseline CPAP (usually 3-4cm above base). Pressure High Flow meter (1-5 L/m) determines PIP or Pressure Assist.

11. Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

11.1. Key performance indicators comprise:

- Nice Neonatal Quality Standards
- NHS Toolkit for High Quality Neonatal Services
- National Neonatal Audit Programme
- NHS Standard Contract for Neonatal Critical Care

Process for Implementation and Monitoring Compliance and Effectiveness

Staff are informed of revised documentation. There is an expectation that staff are responsible to keep updated on any improvements to practice and deliver care accordingly.

Data is collected by use of Badger data base.

Incidents are monitored by the neonatal governance team and neonatal network. Incidents are reported by the Datix system and as applicable to the South West Neonatal Network

Non-adherence is reviewed and action plans made if required. Discussion and reviews occur at Directorate meetings, Governance meetings and Ward meetings. Learning and action plans are cascaded at these meetings and improvements implemented. Key findings and learning points will be disseminated to relevant staff.

12. References

- Amatya, S., Macomber, M., Bhutada, A., Rastogi, D., Rastogi, S. (2017). Sudden versus gradual pressure wean from Nasal CPAP in preterm infants: a randomized controlled trial. [Online] Available at: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5446290/ [Accessed 12/11/2018]
- Boxwell G (2012). Neonatal Intensive Care Nursing Routledge. London
- BAPM. (2015). National Neonatal Critical Care HRGs. [Online] Available at: https://www.bapm.org/resources/national-neonatal-critical-care-hrgs-2015 [Accessed 11/10/2018]
  Carefusion (2018) Taking neonatal noninvasive ventilation to the next level. [Online]. Available at: https://www.vyaire.com/...care/...ventilation/RC_Infant-Flow-SiPAP-BR_EN_INT.pdf. *[Accessed 04/03/2018]*
- Cloherty J and Stark A. (1998). Manual of Neonatal Care. 4th edition. Lippincott – Raven. New York

- Davis PG, Henderson-Smart DJ. (2003). Nasal continuous positive airways pressure immediately after extubation for preventing morbidity in preterm infants. The Cochrane Database of Systematic Reviews 2003, Issue 2. Art. No.: CD000143. DOI: 10.1002/14651858.CD000143.
- Davis PG, Lemyre B, De Paoli AG. (2001). Nasal intermittent positive pressure ventilation (NIPPV) versus nasal continuous positive airway pressure (NCPAP) for preterm neonates after extubation. The Cochrane Database of Systematic Reviews 2001, Issue 3. Art. No.: CD003212. DOI: 10.1002/14651858.CD003212.
- De Paoili A.G., Morley C., Davis P.G. (2003) Nasal CPAP for neonates: what do we know in 2003. Archives of Diseases in Childhood Fetal & Neonatal Edn. 88, F168
- De Paoli AG, Davis PG, Faber B, Morley CJ. (2004) Devices and pressure sources for administration of nasal continuous positive airway pressure (NCPAP) in preterm neonates. Cochrane Library 2 2004.
- EME The Infant Flow System Clinical Training Manual Produced by EME (Electro Medical Equipment Ltd)
- Goggin M (2001). Developments in NCPAP fixation. Journal of Neonatal Nursing. vol 7 no 2
- Gupta, S. & Donn, S. M. Seminars in Fetal and Neonatal Medicine Continuous positive airway pressure: Physiology and comparison of devices. Semin. Fetal Neonatal Med. 21, 204–211 (2016).
- Halliday H McClure B Reid M. (2001). Handbook of Neonatal Intensive Care. 4th edition. WB Saunders Co Ltd
- Ho JJ, Subramaniam P, Henderson-Smart DJ, Davis PG. (2002a). Continuous distending pressure for respiratory distress syndrome in preterm infants. The Cochrane Database of Systematic Reviews 2002, Issue 2. Art. No.: CD002271. DOI: 10.1002/14651858.CD002271.
- Ho JJ, Henderson-Smart DJ, Davis PG. (2002b). Early versus delayed initiation of continuous distending pressure for respiratory distress syndrome in preterm infants. The Cochrane Database of Systematic Reviews 2002, Issue 2. Art. No.: CD002975. DOI: 10.1002/14651858.CD002975.
- Jollye, S and Summers, D. (2010). Chapter 6: Management of Respiratory Disorders cited in: Boxwell, G. (2010). Neonatal Intensive Care Nursing: 2nd Edition. Abington: Routledge.
- Kirpalani H, Millar D, Lemyre B, Yoder BA, Chiu A, and Roberts RS. (2013) A trial comparing noninvasive ventilation strategies in preterm infants. N Engl J Med 369: 611- 620.Lemyre B, Davis PG, De Paoli AG. (2002). Nasal intermittent positive pressure ventilation (NIPPV) versus nasal continuous positive airway pressure (NCPAP) for apnea of prematurity. The Cochrane Database of Systematic Reviews 2002, Issue 1. Art. No.: CD002272. DOI: 10.1002/14651858.CD002272.
- Lamburne S (2014) An assessment tool for infants requiring nasal continuous positive airway pressure. Infant 10 (4) 123-126.
- Lavelle, J., and McCoy, R. (2018). UHL Neonatal Guideline: CPAP Nursing Care. [Online] Available at: https://secure.library.leicestershospitals.nhs.uk/PAGL/Shared%20Documents/Continuous%20Positive%20Airway%20Pressure%20(NCPAP)%20UHL%20Neonatal%20Guideline.pdf [Accessed 30/09/21]

- Levene M, Tudehope D, Thearle M. (2000). Essentials of Neonatal Medicine 3rd edition. Blackwell Science Ltd
- Litchfield S. (1998). Nasal continuous positive airway pressure, does it increase handling in the preterm infant? Journal of Neonatal Nursing. vol 4 no 2 p30-34
- Liverpool Womens NHS Foundation Trust. Guidelines for the use of Nasal CPAP on the Neonatal Unit. [Online] Available at: http://www.liverpoolwomens.nhs.uk/Library/health_professionals/Neonatal_policy_library/CPAP%20on%20the%20Neonatal%20Unit.pdf. [Accessed 07/02/14]
- Mazella M, Bellini C et al (2001). A randomised control study comparing the infant flow driver with nasal continuous positive airway pressure in preterm infants Archives of Disease in Childhood Fetal & Neonatal Edition 85(2) p86-90
- Morley CJ, Davis PG, Doyle LW. (2008) Nasal CPAP or intubation at birth for very preterm infants. N EnglJ Med 2008;358:700-8
- NDDH SCU (2018). Neonatal Care Planning. NDDH.
- Newborn Services Clinical Guideline 2004. Continuous Positive Airways Pressure Guideline. [online]. www.adhb.govt.nz/newborn/Guidelines/Respiratory/CPAP Accessed (07/02/14)
- NHS Forth Valley. Continuous Positive Airway Pressure (CPAP) Nursing Guidelines. [online]. Available at: http://www.nhsforthvalley.com/__documents/qi/ce_guideline_wcdneonatal/continuous-positive-airway-pressure-cpap.pdf. [Accessed 07/02/14]
- Pandit P, Courtney S et al (2001). Work of breathing during constant and variable flow nasal CPAP in preterm neonates. Pediatrics Sept 108(3) p682-685
- Resus Council (UK) (2016). Newborn Life Support (4th edition)
- Singh SD, Clarke P, Bowe L, Robinson MJ, Smith J, Glover K, and Pasquill A (2006) Nasal CPAP weaning of VLBW infants: Is decreasing CPAP pressure or increasing time off the better strategy? Results of a randomised controlled trial. Journal Early Human Development. 83(2) 130-1.
- South Central Neonatal Network Quality Care Group (2012) Guideline framework for CPAP. [online] Available at: http://www.networks.nhs.uk/nhs-networks/thames-valley-wessex-neonatal-network/documents/guidelines/CPAP%20guidline.pdf [Accessed 07/02/14]
- Subramaniam P, Henderson-Smart DJ, Davis PG. (2005).Prophylactic nasal continuous positive airways pressure for preventing morbidity and mortality in very preterm infants. The Cochrane Database of Systematic Reviews 2005, Issue 3. Art. No.: CD001243. DOI: 10.1002/14651858.CD001243.pub2.
- Subramaniam, P., Ho, J.J., Davis, P.G. (2016). Nasal continuous positive airways pressure started immediately after birth for preventing illness and death in very preterm infants. [Online] Available at: https://www.cochrane.org/CD001243/NEONATAL_nasal-continuous-positive-airways-pressure-started-immediately-after-birth-preventing-illness-and [Accessed 11/10/2018]

- Sunil, B., Girish, N., and Manali Bhuyan. (2017). Outcome of preterm babies with respiratory distress syndrome on nasal CPAP. [Online] Available at: http://www.ijpediatrics.com/index.php/ijcp/article/viewFile/888/718 [Accessed 11/10/2018]
- SUPPORT study group.(2010). Early CPAP versus surfactant in extremely preterm infants. N Engl J Med 2010;362:1970-9
- Te Pas A and Wlather F; (2007). A randomised controlled trial of delivery-room respiratory management in very preterm infants Pediatrics 120:322-329
- Yeo H (1998) Nursing the Neonate. Blackwell Science Ltd. Oxford

13. Associated Documentation

14. Appendices

14.1. Appendix One

15. Skin assessment score for pressure areas specific to SiPAP and Ventilated infants

SIGNS	SCORE	ACTION
Nares, mouth, ears, areas under straps, hats, and tubes all appear healthy.	0	No action required. Relieve pressurized areas with cares as able
Slight redness noted around nares, mouth, ears or under straps, hats and tubes. Area appears painful to touch. Some indentation noted.	1	Check for correct size hat and mask Assess and discuss with senior nurse/Dr how to manage .e.g. change from mask to prongs, put gauze to relieve pressure under straps, to change ET tube from oral to nasal, change position Change frequency of assessment Document in the Infant's notes and on the infant's ITU chart.
All of the following evident; Marked indentation, Painful to touch, Tissue breakdown.	2	Call; Registrar/consultant/senior nurse to discuss and change method of care. Remove device ensuring infant's safety. Document as above and complete an incident form. Ensure parents are informed. Dr to refer to plastic surgeons and or medical photographer as appropriate.

(NDDH 2018 From Nursing Care Plan no.7 – Skin Integrity)

Associated Documentation (Optional)

- E.g. related policies